



350 years of rain data digitised
1600 volunteers transcribed and digitised
350 years worth of rain data.
<https://digit.in/jul21-05>



New medical app by Google
Google's dermatology AI app was approved
in the EU but FDA is yet to approve it.
<https://digit.in/jul21-06>



CLEAN DRINKING WATER

Desalination technologies, particularly filtration, has been a surprising focus of research and development.

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bout 785 million people around the world lack a source of clean drinking water, and filtering

seawater is one of the most energy efficient ways of producing clean, potable water. The world is facing a freshwater shortage crisis and one of

the most promising and accessible technologies to tackle this problem is filtration. Filtering water seems like a pretty simple matter. You have the filtration material or membrane, you push the water through it, the contaminants get trapped in the filter, and clean water comes out of the other end. However, researchers do not really understand all the processes that are taking place within the filters on a nanoscale, until now. In desalination plants, reverse osmosis membranes work to filter out the salts and minerals from sea water, producing fresh water using less energy than other approaches that do not use membranes.

Developing novel polymer membranes, as well as nanomaterials have allowed researchers to create smaller and more efficient membranes. Researchers from Korea Institute

of Civil Engineering and Building Technology (KICT) have developed a new method of producing three dimensional hierarchical structures designed with rougher surfaces for better hydrophobicity, using nanofiber membranes produced through electrospinning. These are extremely fine fibers are much smaller than conventional fibers, and thus better at trapping the tiniest of particles and pollutants. The hydrophobic silica gel used to keep storage dry was spun into the fibers in the form of silica aerogel, producing far better filtration membranes than previous approaches using electrospinning. The new filters can continue filtering up to a month, whereas the older filters struggled to keep functioning after 50 hours. Researchers from Kobe university have also developed nanotech filters that sandwiches two two dimensional sheets of graphene oxide on the surface of a porous polymer membrane. The desalination layer is only 50 nanometers in thickness. The research paves the way for future high performance membrane technologies for filtration. Additionally, the gaps between the two dimensional sheets can be managed, and provided with a charge, opening up new use cases for future membrane applications.

According to conventional wisdom, the thicker the polymer membranes used in desalination plants, the less water is produced, but engineers from DuPont found that thicker membranes were actually filtering more water, and funded a study to find out what was going on. Chemical and civil engineers teamed up together and used multimodal electron microscopy to probe the desalination membranes used in saltwater treatment plants, and found that the membranes had inconsistencies in mass and densities. They used simulations to check how water flowed through the filters to predict how efficiently water could be cleaned in these filters. If you have seen a badminton racket or a